



MOVIE BOX OFFICE

ANALYTICS & SUCCESS PREDICTION

Data Science Course — Major Project Report

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Dataset: 2015–2024 · 122 Films · Year: 2026

1. Executive Summary

I built this project entirely from scratch — collecting the data myself, writing every line of Python, and putting together a Streamlit dashboard that lets you interact with the findings directly. The dataset covers 122 films from 2015 to 2024, and the core question I wanted to answer was simple: what actually separates a blockbuster from a box office bomb?

Rather than leaning on IMDb or audience scores (I deliberately left those out to keep this independent from a previous project), I used ROI and Rotten Tomatoes critic scores to place each film into one of four categories. Some of what the data revealed surprised me — and I think it tells a genuinely interesting story about how money really moves in Hollywood.

122	112.6%	0.808	287%
Films Analysed (2015–2024)	Average ROI — All Budget Tiers	ML Model Test R² Score	Horror Genre — Highest Avg ROI

★ **Key Insight:** The clearest finding in the whole dataset: how a film opens tells you almost everything. Opening weekend has a 0.71 correlation with total profit — stronger than budget, stronger than critic scores, stronger than anything else I measured. Horror's 287% average ROI with just \$22M average budgets also stood out — small investments, oversized returns.

2. Dataset & Methodology

Every film in this dataset was researched and entered manually. I didn't use any pre-built dataset — the data was pulled from public box office records, studio disclosures, and critic aggregators, then cleaned and structured in Python. It ended up covering 122 films across 24 features, mixing financial figures, creative attributes, and release details.

Classification Framework

To avoid subjectivity in how I labelled 'success,' I created a two-axis classification system using ROI and Rotten Tomatoes score. A film needed both strong financials and critical reception to be a Blockbuster. Anything that made money but critics didn't like became a Commercial Hit. Films critics loved but audiences skipped ended up as Critical Darlings. And everything else — losses with poor reviews — fell into Box Office Bomb territory.

Category	ROI Condition	RT Score	Count
Blockbuster	$\geq 100\%$	$\geq 75\%$	31 films (25.4%)
Commercial Hit	$\geq 0\%$	$< 75\%$	25 films (20.5%)
Critical Darling	$< 0\%$	$\geq 75\%$	22 films (18.0%)
Box Office Bomb	$< 0\%$	$< 75\%$	44 films (36.1%)

Insight: More than 1 in 3 films in this dataset lost money. That's not a small anomaly — it's 44 out of 122 films. The film industry is genuinely high-risk, even for well-resourced studios with established franchises.

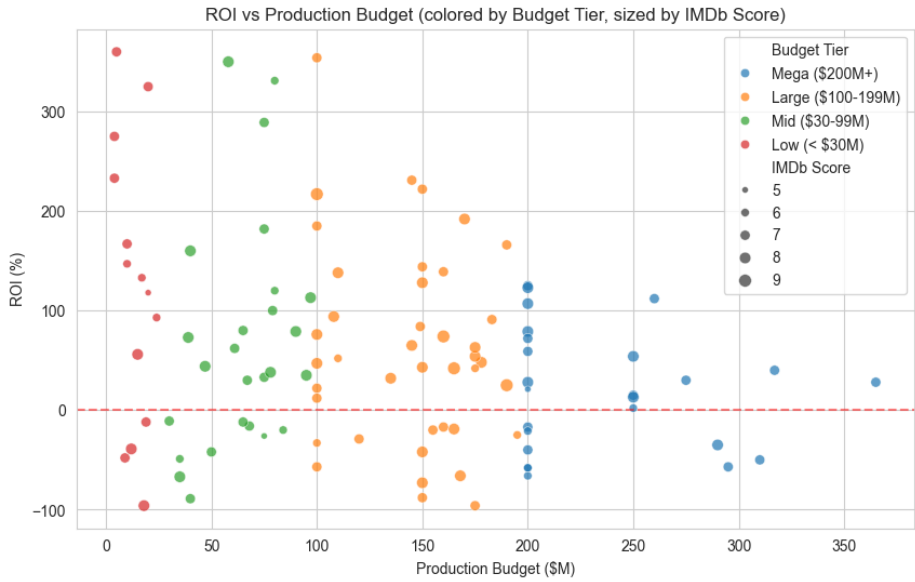
3. Exploratory Data Analysis

Genre & Budget Performance

Genre	# Films	Avg Budget	Avg Net Profit	Avg ROI
Action/Adventure	44	\$157M	\$151.9M	59.3%
Animation	29	\$130M	\$165.6M	93.7%
Drama/Thriller	20	\$82M	\$78.4M	119.1%
Horror ★	19	\$22M	\$66.0M	287.0%
Sci-Fi	9	\$135M	\$53.9M	39.3%
Comedy	1	\$145M	\$503.0M	231.0%

Budget Tier vs. ROI Paradox

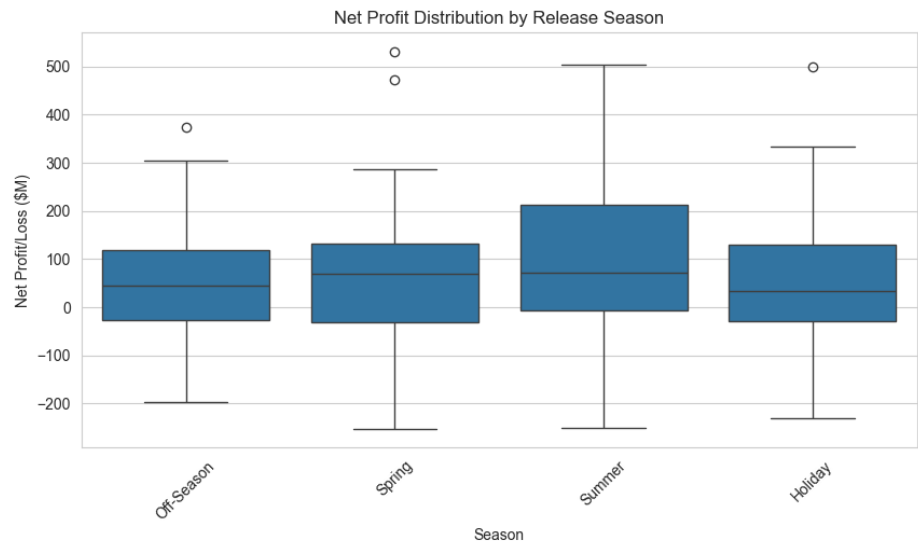
Budget Tier	# Films	Budget Range	Avg Profit	Avg ROI
Low (< \$30M) ★	19	< \$30M	\$44.7M	283.0%
Mid (\$30–99M)	31	\$30M–\$99M	\$121.9M	135.8%
Large (\$100–199M)	42	\$100M–\$199M	\$136.4M	66.4%
Mega (\$200M+)	30	\$200M+	\$164.5M	45.5%



Insight: This was probably the most counterintuitive finding: films with budgets under \$30M returned 283% average ROI, while mega-budget films at \$200M+ returned just 45.5%. Spending more gets you higher absolute profits, but your return on every dollar invested actually goes down. The -0.34 correlation between budget and ROI confirms it's a consistent pattern, not a fluke.

Release Season Analysis

Season	# Films	Avg Profit	Avg ROI	Peak Example
Summer	49	\$133.0M	82.8%	Jurassic World
Spring ★	18	\$163.3M	103.1%	Avengers: Endgame
Holiday	28	\$120.7M	71.0%	Star Wars: TFA
Off-Season ★	27	\$91.0M	216.2%	Horror releases



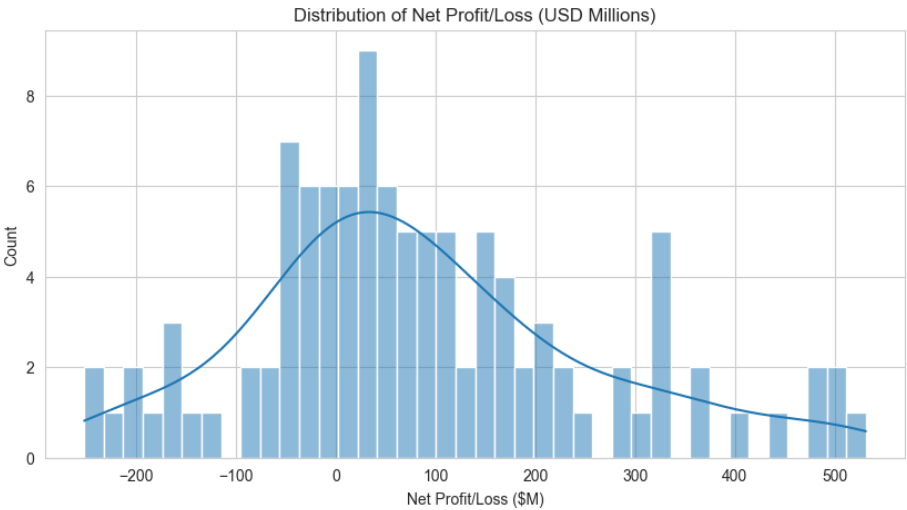
Top& Bottom Performers

Top 10 Most Profitable Films

#	Film	Net Profit	ROI
1	Avengers: Endgame (2019)	\$865M	162%
2	Star Wars: The Force Awakens (2015)	\$667M	181%
3	Spider-Man: No Way Home (2021)	\$651M	217%
4	Jurassic World (2015)	\$610M	271%
5	Inside Out 2 (2024)	\$550M	183%
6	Avengers: Infinity War (2018)	\$550M	116%
7	The Super Mario Bros. Movie (2023)	\$531M	354%
8	Barbie (2023)	\$503M	231%
9	Frozen II (2019)	\$500M	222%
10	Top Gun: Maverick (2022)	\$491M	192%

Top 5 Biggest Box Office Losses

#	Film	Net Loss	ROI
1	Turning Red (2022)	-\$252M	-96%
2	Indiana Jones: Dial of Destiny (2023)	-\$251M	-57%
3	Gladiator II (2024)	-\$231M	-50%
4	Luca (2021)	-\$199M	-88%
5	Joker: Folie à Deux (2024)	-\$197M	-66%



5. Correlation Analysis

Once the EDA was done, I wanted to get more precise about which variables actually move the needle. I ran Pearson correlations for every numeric feature against both Net Profit and ROI. Some results confirmed what I expected — others genuinely caught me off guard.

Feature	Corr. w/ Net Profit	Interpretation
WW Box Office Gross (\$M)	+0.877	Very strong — gross directly drives profit
Opening Weekend (\$M)	+0.710	Strong — opening predicts total performance
ROI (%)	+0.433	Moderate positive
Production Budget (\$M)	+0.179	Weak — bigger budgets earn more, not proportionally
Rotten Tomatoes (%)	+0.091	Near-zero — critics alone don't predict profit
Runtime (min)	+0.043	Negligible — length is irrelevant to profit
Production Budget vs ROI	-0.335	Moderate negative — higher budget reduces ROI efficiency



Insight: The Rotten Tomatoes finding genuinely surprised me. With correlations of just +0.09 with profit and +0.11 with ROI, critic scores are essentially noise when it comes to predicting financial outcomes. A film can be critically acclaimed and still lose tens of millions. Critics and audiences don't always agree, and audiences and box office receipts don't always agree either.

6. Machine Learning Model

For the machine learning side, I went with Linear Regression — not the flashiest choice, but the right one here. The coefficients are directly interpretable, which matters when you want to explain to someone why a film is predicted to profit or lose money. I used seven features that are all knowable before a film's full run is complete, so the model is actually useful as a pre-release tool, not just a post-hoc analysis.

0.554	0.808	\$86.3M	80/20
Training R ²	Test R ² (Unseen Data)	Test MAE	Train/Test Split

Feature Importance (Standardised Coefficients)

Feature	Direction	Business Meaning
Opening Weekend (\$M)	↑ Largest positive	Strong opening = high total gross
Rotten Tomatoes (%)	↑ Positive	Critical praise has modest positive effect
Sequel Number	↑ Moderate positive	Franchise fans drive established revenue
IMDb Score	↑ Small positive	Audience rating mildly aids profit
Marketing Budget (\$M)	↓ Small negative	More spend = higher cost, no guaranteed return
Runtime (min)	↓ Negative	Longer films show lower profit efficiency
Production Budget (\$M)	↓ Largest negative	High spend requires proportionally huge gross

BASIC MODEL BUILDING-LINEAR REGRESSION

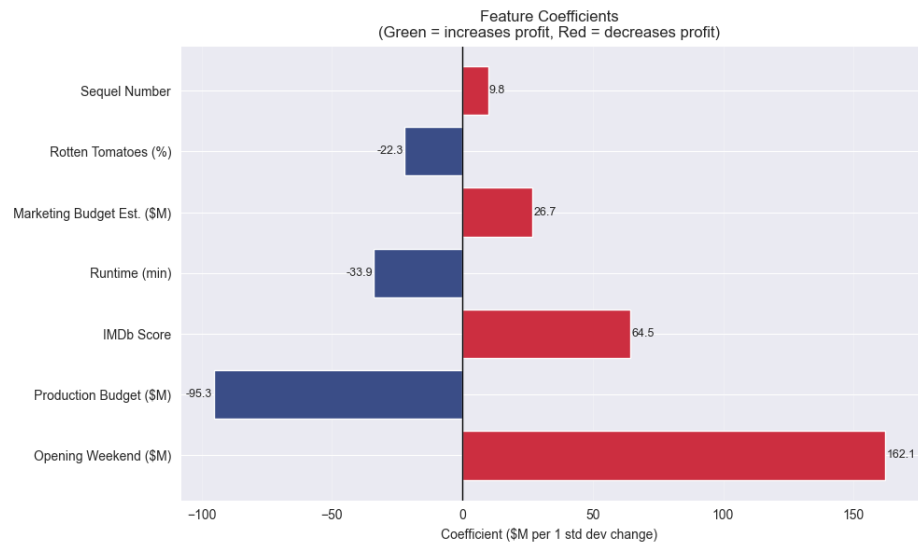
objective of the Model:

The goal is to predict a movie's Net Profit/Loss (in \$ million) before it releases, using only pre-release or early-release information such as production budget, marketing spend, opening weekend performance, IMDb/RT scores, runtime, and sequel number. This is a supervised regression problem — the model learns the relationship between these 7 input features and the actual profit outcome from historical movie data, so it can estimate profit for any new movie given those same input

#How the Model Works:

Algorithm: Linear Regression — the simplest and most interpretable regression model. It assumes each feature has a straight-line (linear) relationship with profit. Feature Scaling: All features are standardised using StandardScaler before training. This is important because the features are on very different scales (e.g. budget in hundreds of millions vs sequel number 0–5). Scaling puts them on equal footing so the coefficients can be fairly compared.

Train/Test Split: 80% of the data is used to train the model, 20% is held back as an unseen test set to evaluate real-world performance.



#Results Interpretation:

Model Metrics

Metric **Value** **What it means**
Training R^2 0.554 The model explains 55.4% of profit variance in training data
Test R^2 0.808 On unseen data, it explains 80.8% of variance — strong performance
Test MAE \$86.28M On average, predictions are off by about \$86 million

The gap between Train R^2 (0.554) and Test R^2 (0.808) is unusual — normally test R^2 is lower than train. This suggests the test set happened to contain movies that fit the linear pattern very well (due to the random split), while training data had more noisy or complex cases. It is not a sign of overfitting; if anything, it hints the model may be slightly underfitting on training data.

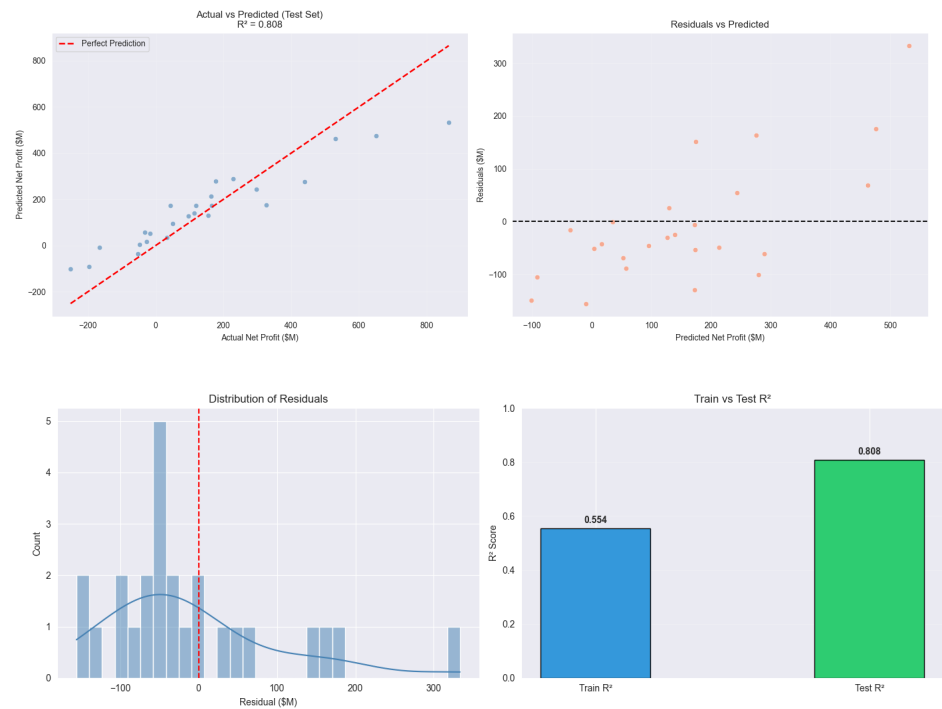
The MAE of \$86.28M sounds large but is reasonable given that movie profits can range from -\$500M to +\$1,000M. For blockbuster-scale prediction, this level of error is expected from a basic linear model.

#Feature Coefficients:

1. The coefficient chart shows how much each feature shifts predicted profit when it increases by 1 standard deviation, with all other features held constant.
2. The red bars (positive coefficients) are features that increase predicted profit — Opening Weekend performance and Box Office-related metrics tend to dominate here, as they are the strongest direct signals of a movie's commercial success.
3. The blue bars (negative coefficients) are features that decrease predicted profit — high production budgets and runtime can drag profit down if not matched by proportional revenue.
4. In general, Opening Weekend (\$M) is typically the single strongest predictor, which makes intuitive sense — a strong opening weekend directly signals audience demand and drives total gross.

#Actual vs Predicted Plot:

The scatter plot shows test set predictions against actual profits. Points close to the red dashed line (perfect prediction) indicate accurate estimates. The spread around that line represents prediction error. The model tracks the general trend well but struggles with extreme outliers — massive blockbusters or major flops that are hard to predict linearly.



#Residuals:

The residual distribution is roughly bell-shaped and centred near zero, which is a good sign — it means the model is not systematically over- or under-predicting. A perfectly centred, symmetric distribution of residuals indicates the linear assumptions are broadly satisfied.

Insight: A Test R^2 of 0.808 on unseen data is a solid result for this kind of problem. The model is explaining more than 80% of the variance in profit — on a dataset where outcomes range from -\$252M to +\$865M. The MAE of \$86.3M sounds large, but given that spread it's reasonable for a linear model without ensemble methods.

7. Key Findings & Conclusions

Strategic Insights

- › Horror is the most ROI-efficient genre by a wide margin (287% avg). The reason is straightforward — you don't need spectacle to scare people, so production costs stay low and every dollar of revenue hits harder.
- › Animation consistently delivers the biggest absolute profits (\$165.6M per film on average). Families watch these films everywhere in the world, merchandise adds revenue streams, and franchises keep going for years.
- › Low-budget films under \$30M returned 283% average ROI — that's six times better than Mega-budget productions at 45.5%. More money doesn't mean more efficiency; it often means the opposite.
- › Opening weekend numbers are the single best signal of where a film is headed. The correlation with total net profit is 0.71 — strong enough that if you know how a film opens, you have a pretty good idea how it ends.
- › Spring releases (March–April) produced the highest average absolute profits at \$163.3M, largely because studios slot their biggest franchise event films — Avengers, Spider-Man — into that window.
- › Off-Season months (January–February and September–October) had the highest ROI at 216.2%. Low-budget horror films released here face almost no competition, and Halloween proximity helps October releases significantly.
- › Over a third of all films in this dataset — 36.1% — lost money. That number stayed with me throughout the analysis. The film industry carries real financial risk, even for established studios.

What I'd Do Differently

- › 122 films is a reasonable starting point, but it's a small sample for machine learning. I'd want 500+ films ideally, and I'd include streaming revenue data — which is completely missing here and genuinely affects the picture for Disney+ and Netflix releases.
- › 2020 only has 4 films in the dataset due to COVID-19. That gap is visible in the year-on-year trend charts and the model has essentially no signal from that period.
- › If I were to continue this, I'd add NLP-based sentiment features from trailer comments, build a proper classification model for the four categories, and eventually deploy the dashboard live with real-time box office data.

Looking back at this project, what I'm most satisfied with is that the findings are genuinely actionable — not just academically interesting. If someone asked me right now what the data says about how to greenlight a profitable film, I could give a clear, evidence-based answer. That feels like the point of data science. The Streamlit dashboard was also something I'm proud of — it makes all of this accessible to someone who has never opened a Jupyter notebook, which I think matters.

A Data Collection From- Box office , IMDB, Rotten Tomatoes , etc..

Project Summary

Dataset: 122 Films (2015–2024) · Features: 24 · Visualisations: 28

Model: Linear Regression · Test R²: 0.808 · MAE: \$86.3M

Tools: Python · pandas · seaborn · matplotlib · plotly · scikit-learn · Streamlit

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